

Chapter 1

Introduction to Databases & E-R Model

- Q.1 What is Database Management System? Explain the primary goals of DBMS.
- Q.2 What is Database Management System? Give its applications.
Explain Traditional File Processing System.
- Q.3 OR
Compare Database Management System and Traditional File Based System.
- Q.4 What is Database Management System? Explain the purpose of DBMS.
What is the view of data? Explain different levels of data abstraction.
- Q.5 OR
Explain different levels of data abstraction with example.
- Q.6 Explain the terms - i) View of data ii) Instances and Schema
- Q.7 Explain Data models and its different categories.
- Q.8 Explain two-tier and three-tier database architecture.
- Q.9 What is E-R model? Explain basic components of E-R model.
- Q.10 What is attribute? List & explain different types of attributes with example.
- Q.11 List & explain different types of constraints in detail with example.
- Q.12 What is key? List & explain different types of keys with example.
- Q.13 What is E-R diagram? Explain the components of E-R diagram.
- Q.14 What is mapping cardinality? How to denote mapping cardinality in E-R diagram?
Write a note on Weak-entity set & Strong-entity set
- Q.15 OR
Differentiate between weak entity set and strong entity set.
- Q.16 How to reduce E-R diagrams into relational schemas?
- Q.17 Explain extended E-R features with example.

Chapter 2

Relational Model & SQL

- Q.1 What is SQL? List and explain the commonly used data types in SQL.
- Q.2 What are different categories of SQL Commands? Explain with example.
- Q.3 What is SQL? Explain DDL statements.
- Q.4 What is SQL? Explain DML statements.
- Q.5 What is SQL? Explain DCL statements.
- Q.6 What is SQL? Explain TCL statements.
- Q.7 What is the use of Data Definition Language? Explain SQL Constraints with example.
- Q.8 Explain different SQL Constraints with example.
- Q.9 Differentiate between Grant and Revoke.
- Q.10 Explain different string operations performed in SQL.
- Q.11 Explain with an example use of the ORDER BY clause in a query.
- Q.12 How does the GROUP BY clause work in SQL? Explain with example.
- Q.13 What is the purpose of the HAVING clause in SQL? Differentiate between WHERE and HAVING clauses.
- Q.14 Explain with an example use of the WHERE clause in a query.
- Q.15 What are set operations in SQL? Explain with example.
- Q.16 What are aggregate functions in SQL? Explain with example.
- Q.17 What are nested queries in SQL?
- Q.18 What are different operations used in modification of database?
- Q.19 Explain different types of joins in SQL.
- Q.20 What is a view in SQL? Explain the benefits of using views in a database system.

Chapter 3

Relational Database Design

What is integrity constraints? List & explain its types.

Q.1 OR

Explain different types of integrity constraints.

What is Functional Dependency? Explain in detail.

OR

Q.2 Define functional dependency? What are the different types of functional dependencies?

OR

Explain different types of functional dependencies.

Q.3 What is Functional Dependency? Explain its advantages.

Q.4 What is Normalization? Explain the purpose of normalization in database design?

Q.5 What is Normalization? Explain different types of anomalies in database.

Q.6 Explain First Normal Form (1NF) and its significance.

Q.7 Describe Second Normal Form (2NF) and how it differs from 1NF.

Q.8 Discuss Third Normal Form (3NF) and its importance in database design.

Q.9 What is Boyce-Codd Normal Form (BCNF) and when is it applicable?

Q.10 Explain Fourth Normal Form (4NF) and Fifth Normal Form (5NF) and their significance in database normalization.

Chapter 4

Data Storage & Indexing

- Q.1 List & explain different types of physical storage media available.
- Q.2 What is RAID? Explain different levels of RAID.
What are different ways to represent the records?
- Q.3 OR
Explain Fixed-Length & Variable-Length Records.
How to store variable length records within the block?
- Q.4 OR
How slotted page structure is used to store variable length records within the block?
- Q.5 What are different ways to organize records in files?
- Q.6 Explain Sequential file organization.
- Q.7 Explain Clustering file organization.
- Q.8 What is Data Dictionary? Explain Data Dictionary Storage.
How buffers are used to minimize the number of block transfers between the disk and memory?
- Q.9 OR
Explain Buffer Manager.
What is Buffer Manager? Which techniques used by buffer manager to serve database system in well manner?
- Q.10
- Q.11 What is indexing? Explain different types of indexing.
- Q.12 What are the different factors to evaluate indexing techniques?
- Q.13 What is indexing? Explain Ordered indexing.
What is indexing? Explain different types of Ordered indices.
- Q.14 OR
What is indexing? Explain Sparse index & Dense index.
- Q.15 What is multilevel indexing? What is the need for multilevel indices?
- Q.16 What is hashing? Explain the different components in hashing.

- Q.17 What is hash function? How it works?
- Q.18 What is hashing? Explain types of hashing.
- Q.19 What is hashing? Explain Static Hashing.
- Q.20 What is hashing? Explain Dynamic Hashing.
- Q.21 How to resolve the problem of bucket overflow in static hashing?

Chapter 5

Concurrency Control & Crash Recovery

- Q.1 What is transaction? List & Explain the properties preserved by it.
- Q.2 Explain ACID Properties in detail.
- Q.3 What is transaction? Explain the states of transaction with suitable state diagram.
- Q.4 What is transaction? Give the operations by which transactions access the data.
- Q.5 Explain shadow database scheme with neat diagram.
What is schedule? Explain the types of schedules.
- Q.6 OR
What is schedule? Explain serial & non-serial schedule.
- Q.7 What is serializability? Discuss its types.
- Q.8 When the schedules are view equivalent?
- Q.9 Explain view serializability with example.
- Q.10 Explain conflict serializability with example.
- Q.11 What is recoverability? Explain different schedules scenario in recoverability.
- Q.12 What is serializability? How to test the serializability?
- Q.13 What is concurrency control? Explain Lock Based protocol.
- Q.14 Discuss two-phase locking protocol.
- Q.15 Explain Timestamp based protocol.
- Q.16 Explain Validation based protocol.
- Q.17 What do you mean by data recovery? What are the types of failures?
- Q.18 Explain log based recovery scheme.
- Q.19 Explain the approaches used in log based recovery scheme.
- Q.20 Explain the concept of checkpoint with example.
- Q.21 Explain Shadow Paging.

Chapter 6

Database Security & Authorization

- Q.1 What are the threats to database ?
- Q.2 What are different control measures used to provide security of data in databases ?
- Q.3 What are the different factors to classify data as sensitive ?
- Q.4 What factors need to be considered before deciding whether access is safe to reveal the data?
- Q.5 Explain discretionary access control (DAC) along with granting and revoking of privileges.
- Q.6 What are the types of discretionary privileges? Explain in detail.
- Q.7 How privileges are propagated in DAC? Explain with example.
- Q.8 Explain mandatory access control (MAC).
- Compare Discretionary Access Control and Mandatory Access Control.
- OR
- Q.9 Differentiate between Discretionary Access Control and Mandatory Access Control.
- Q.10 Explain Role Based Access Control.
- Q.11 What is SQL Injection? Explain different types of attacks on databases.
- Q.12 What is SQL Injection? Explain different types of injection attacks.
- Q.13 What is SQL Injection? Explain risk associated with SQL Injection.
- Q.14 What is SQL Injection? Explain protection techniques against SQL injection.